Fiscal Year:	FY 2012	**Task Last Updated:**	FY 11/02/2011
PI Name:	Turker, Mitchell Ph.D.		
Project Title:	The Relation Between Mutagenesis and Genomic Instability After Particle Exposure In Vivo		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Space Biology Special Category:	None		
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Zip Code:	97239-3011	**Congressional District:**	1
Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
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COI Name (Institution):	Kronenberg, Amy (Lawrence Berkeley National Laboratory)		
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Task Description:	Highly energetic charged particles in space environments are hazardous because they cause DNA damage, which leads to mutation and loss of expression of critical genes that normally prevent cells from becoming cancerous. Most charged particle-induced mutations at autosomal loci span large regions of an affected chromosome resulting in loss of heterozygosity (LOH) for these regions. The proposed work has three interrelated goals. The first is to document the induction of mutational processes leading to LOH in two tissues, lung and spleen. The mutagenic effects of three charged particles will be tested: 1 GeV/amu Fe ions, 1 GeV/amu Ti ions, and 1 GeV protons to address the impact of ionization density on mutagenesis. The second goal is to demonstrate that LOH occurs on multiple chromosomes in the induced mutant cells, which would suggest the development of genomic instability. The third goal is to demonstrate directly with novel color markers that genomic instability characterized by high levels of LOH develops in the progeny of cells exposed in vivo. The basis for this approach is that selection of cells with LOH type mutations will provide the "bait" to		

	"fish out" cells exhibiting charged particle-induced genomic instability. Successful completion of the proposed work will identify those types of mutational events that reflect charged particle-induced genomic instability, and the role of ionization density in the development of these unstable cells in apparently normal tissues. The results will be useful in assessing cancer risks from space travel in two tissues that are prone to radiogenic cancers in humans.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Overview. The objective of the ongoing project is to determine the effects of accelerated particles found in the space environment on cells in intact organs. These particles contain nuclei stripped of their electrons. The three particles under study are iron (Fe) ions, titanium (Ti) ions, and protons (a hydrogen nucleus). To determine the effects in the intact organs, live mice are exposed to the accelerated particles at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). The mice are then returned to Oregon Health Science University (OHSU) in Portland, OR for testing approximately 4-5 months after exposure at BNL. Radiation effects are measured in the cells from three tissues: lung, spleen, and kidney. We are specifically funded for the lung and spleen cells, but have added kidney for some assays because we have shown in over 8 years of prior work that the cells of this organ are sensitive to accelerated particles. Thus, the kidney serves as a good control that our assays are working and that the exposures at BNL were effective. Two end points are used. The first is to determine the extent of overall cellular harm (toxicity) from the particle exposure. The second is to determine the extent of harm to the cells' genetic material (i.e., DNA). Mice with a specific genetic background are bred for these studies. The work below will describe the results from our first set of experiments in which the mice were exposed to 200 cGy of Fe ions, 140 cGy of Ti ions, or 400 cGym protons. Unexposed mice are used as controls. These relatively high doses were chosen to help inform work at lower doses, which are more representative of those astronauts would be exposed to during prolonged space travel. We are currently setting up assays for mice exposed to 50 cGy Fe ions or 40 cGy Ti ions and in November 2011 we will expose mice at BNL to 25 cGy Fe ions or 100 cGy protons. The total number mice shipped to BNL to date for irradiation is 223. Cellular harm. The toxic effects from radiation exposure can be determined with a "cloning assay." For this assay, each lung or kidney is treated with enzymes to separate the cells, which are then plated in culture to determine how many cells can grow into a clone. A clone is the product of a single cell that can divide enough times to produce 100 or more progeny cells. The relative number of cells that can grow into clones provides the cloning efficiency for each organ. The spleen can be disrupted manually to create a single cell suspension for the same type of assay. Radiation effects are demonstrated by showing that the cloning efficiency is reduced for cells from organs exposed to radiation in the intact mouse compared with cells from the same organs obtained from non-irradiated mice. Cloning efficiencies were measured for lungs, kidneys, and spleens at 4-5 months after the mice were exposed to Fe ions, Ti ions, or protons. For the lungs, the results showed a 55% reduction in cloning efficiency after Fe ion exposure and 43% decreases after Ti ion and proton exposures. For the kidneys, the results showed an approximately 40% decrease after Fe ion exposure and 30% decreases after Ti ion and proton exposure. All decreases in cloning efficiencies for the lung and kidneys were statistically significant. For the spleen, the results showed a 40% decrease in cloning efficiency after proton exposure, but only approximately 25% decreases after Fe ion or Ti ion exposure. Only the proton caused reduction in cloning efficiency was deemed statistically significant. In total, these results provide evidence that between 25% to 55% of the cells from different mouse organs had long-term impairment from radiation exposure, as evidenced by reduced cloning efficiencies. The level of impairment was influenced by particle, dose and cell type characteristics. DNA harm. DNA is the genetic material within each cell, and DNA damage has been linked to several disease processes including most notably cancer. DNA damage can cause mutations, which are small or large alterations in the DNA sequence. We use mice that have one good copy and one bad copy of a gene called Aprt to measure DNA harm. The important point about the Aprt mice for this discussion is that methods are available to separate cells that have lost the good Aprt copy from those that still retain the good copy. The cells that lose the good copy, which are called Aprt mutants, are present in the non-irradiated mice at frequencies of 1 in 100,000 cells for the spleen and 1 in 10,000 cells for the kidneys and lungs. In other words the mutant cells are quite rare, but we have methods to find them. Radiation can increase the Aprt mutant frequency because it causes DNA damage, which occasionally causes loss of the Aprt gene. The Aprt mutant frequencies increased with statistical significance for the cells obtained from lungs, kidneys, and spleens of mice exposed to Fe ions. The mutant frequencies also increased with statistical significance for the kidney cells exposed to Ti ions or protons. Two-fold increases in mutant frequencies were observed for lungs exposed to Ti ions or protons, but these failed to achieve statistical significance. This could change as more data from control mice are added to the analysis over the next year. The mutant frequencies did not increase for the spleen cells exposed to Ti ions or protons. DNA harm can also be demonstrated by directly examining the DNA molecules from mutant cells for changes on the chromosome that contains the good copy of Aprt. We have obtained DNA from Aprt mutants for most of the conditions described above for this analysis, which will be conducted in the upcoming year. We can also use special probes that allow us to directly visualize this mouse chromosome in mutant cells, and have collected mutant cells from the kidneys and lungs. During the last year we showed that Fe ions and protons caused significant chromosome breaks in kidney mutants. We are currently working on a protocol for collecting Aprt mutant spleen cells for this purpose. Summary. We have successfully demonstrated cellular and DNA harm in the cells of living mice exposed to the types of accelerated particles found in the space environment. This work required us to develop new assays in the laboratory for culturing spleen T cells and lung cells. The lung cell assay that we developed is, to the best of our knowledge, unique to our laboratory. Likewise the kidney cell assay is unique to our laboratory. Work in the coming year will look more closely at DNA damage caused by the accelerated particles, including lower doses of these particles, for both the cellular and DNA harm from radiation.
Bibliography Type:	Description: (Last Updated: 11/01/2023)

Articles in Peer-reviewed Journals	Dan C, Grygoryev, D Sandfort K, Connolly M, Cross B, Lasarev M, Kronenberg A, Turker MS. "Marked aneuploidy and loss of multiple chromosomes are common in autosomal mutants isolated from normal mouse kidney epithelium." Genes Chromosomes Cancer. 2011 Apr;50(4):239-49. Epub 2011 Jan 19. PubMed PMID: 21254298 ; http://dx.doi.org/10.1002/gcc.20849 , Apr-2011